

ECM3401 Project Logbook

Student Name: Dylan Hough

Project Name: Deep Learning for Guitar Tablature Transcription from Monophonic Audio

Internal Supervisor: Dr Alzubi, Saif

External Supervisor: (if any)

Log of Activities

Date	Key activities	Outcomes	Comments/Action plans
23/09/25	Supervisor meeting, discussing the project idea	New ideas for the project, made aware of deadlines to be aware of and thoughts about feasibility of the project.	After the meeting I should focus on reading relevant literature for the project.
09/10/25	Supervisor meeting	Discussion about what I have learnt in the literature I have read, and thoughts about locating useful datasets	More reading for the literature review.
11/10/25	Promising list of datasets I could use	A list of datasets that could be helpful in the project implication and discussed in the literature review	Dadagp and synth tab look like useful datasets for the amount of data they have
12/10/25	Started the literature review	Drafting a title abstract and introduction	Got the base of the 3 parts done may change once I get deeper into the document
16/10/25	Supervisor meeting	Discussion of datasets I have found and how I may use them. Also mentioned what I have written	Data looks good but searching for more would aid the project
21/10/25	Lit review	Finished a draft of the main part of the lit review	Start the project specification
24/10/25	Started project specification	Wrote about aims and objectives and discussed datasets that will be used	Work more on the specification and find more literature swell
06/11/25	Supervisor meeting	Suggestion of a gantt chart	For the project specification I should add a gantt chart
10/11/25	Finished first draft of lit review/project plan	Completed a project timeline section with, expected outcomes, gantt chart, evaluation criteria, limitations, future evaluation and conclusion. I sent the draft over to my supervisor so I can receive feedback	When my supervisor comes back with his feedback it's worth adding in anything he suggests will benefit the document.
12/11/25	Supervisor replied to lit review with feedback	My supervisor suggested I watch my phrasing on certain sentences in the document as it could be suggesting I'm doing something I am not. Also some general feedback about positioning of certain sections as well as adding in a "Risks and ethical considerations" section into the document	I'll revise the document and add any changes my supervisor suggests and make any amendments to produce a final draft ready to be submitted

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13/11/25	Supervisor meeting	Discussion about the feedback produced and further deadlines after the lit review/project plan	Take note of what I should be working on next for the project as a whole in terms of when the deadlines are
18/11/25	Submitted lit review		I will start working on the next part of the project according to the Gantt chart I made
14/01/26	Supervisor meeting	Discussion of what I have achieved over the autumn break and what I plan to do next	For the project I'm going to start manipulating the datasets I have found
29/01/26	Supervisor meeting	Discussion of the dataset work I've been doing with dadagp and synthtab	Further work on the datasets needs to be looked at
7/02/26	Dataset changes	Training using primarily synthetic audio seems to not produce results well for real guitar audio	I should try and find new datasets and primarily focus the training on real guitar playing notes
13/02/26	Promising datasets found and manipulated	Combined datasets such as idmt-smt into a folder format where all the individual notes on each string and fret are split into separate files	Using functions to augment each data point to increase the amount of data to train on
20/02/26	Supervisor meeting	Highlighted the augmentation feature that would increase data. Supervisor suggested I start training soon	Due to time constraints improving the data I have wouldn't be worth it so I will start making the training mechanism.
26/02/26	Supervisor meeting	Reviewing the training pipeline and resolving issues with the code before beginning training	Try to get some progress on the training
02/03/26	Started feature extraction	Extracted mel spectrograms and additional audio features from the dataset ready for model input	Begin building and training the model
03/03/26	Built and trained initial model	First model trained, identified that string accuracy was poor on notes sharing the same pitch	Redesign the model to improve string prediction accuracy
05/03/26	Supervisor meeting	Discussed issues with the model and its low accuracy.	Try improving the model architecture to increase accuracy
14/03/26	Improved model trained	Retrained with a redesigned architecture, significant improvement in accuracy across string and fret predictions	Begin building the note capture program and web interface
16/03/26	Started note capture program	Built a program to detect and capture individual notes from live guitar audio using gate-based onset detection	Test the note capture with the trained model and begin the web interface
18/03/26	Started web interface	Built a browser based interface that takes live guitar input, runs it through the note capture and model, and displays the predicted tablature	Test the full system end to end and start working on the poster

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19/03/26	Supervisor meeting	Discussed success with the program, mentioned I will work on the poster	Work on the poster
21/03/26	Started working on poster	Drafted the poster layout covering the project motivation, methodology, system design and results	Refine the poster and run full model evaluation
22/03/26	Model evaluation and poster finalised	Ran full evaluation on the test set, results show strong accuracy for in-session recordings. Poster completed and ready to submit	Submit poster and begin writing the final report
25/03/26	Submitted poster	Poster summarising the project, methodology and results submitted	Continue writing the final report
02/04/26	Report writing	Introduction, background, methodology and implementation sections drafted	Write evaluation, discussion and conclusion sections
10/04/26	Report writing continued	Evaluation and discussion sections complete, full draft ready	Send draft to supervisor for feedback
27/04/26	Supervisor gave feedback on report	Supervisor suggested minor changes to wording and structure	Apply feedback, proof read and submit the final report
28/04/26	Final report submitted	All changes applied, report proof read and submitted	